

Appendix

Appendix A

An important issue concerning the core explanatory variable in this study is the selection of the industry level when identifying the entry of R&D center foreign firms at the city-industry level. The choice of industry level impacts the interpretation of technology spillover effects from R&D center foreign firms. If the industry is classified as a national economic industry classification at the 2-digit level (hereinafter referred to as gb2 industry), the underlying assumption is that R&D center foreign firms only produce technology spillovers to domestic firms within the same gb2 industry, with no spillovers between different gb2 industries; however, spillovers may occur within the same gb2 industry between different 3-digit national economic industry classifications (hereinafter referred to as gb3 industries) and between different gb4 industries. Similarly, if the industry is classified at the 3-digit level (gb3 industry), the assumption is that R&D center foreign firms only generate spillovers to domestic firms within the same gb3 industry, with no spillovers between different gb3 industries.

In this study, the entry of R&D center foreign firms is identified at the city-gb2 industry level, meaning that the industry level in the core explanatory variable is gb2. Therefore, the treatment group consists of cities-gb2 industries with R&D center foreign firm entry, while the control group consists of cities-gb2 industries without R&D center foreign firm entry. The results remain robust when identifying R&D center foreign firm entry at the city-gb3 and city-gb4 industry levels, as demonstrated in the section 6.3. We provide an explanation of R&D center foreign firms' entry at different industry levels in Appendix Table A1.

Appendix Table A1: An explanation of R&D center foreign firms entry at different industry levels

gb3 industry	gb3 code	gb4 industry	gb4 code	gb2 entry	gb3 entry	gb4 entry
Automobile Manufacturing	361	Manufacturing of Complete Vehicles for Gasoline and Diesel Cars	3611	1	1	1
Automobile Manufacturing	361	Manufacturing of Complete Vehicles for New Energy Vehicles	3612	1	1	0
	362	Manufacturing of Automotive Engines	3620	1	0	0
	363	Manufacturing of Modified Vehicles	3630	1	0	0
	364	Manufacturing of Low-Speed Vehicles	3640	1	0	0
	365	Manufacturing of Electric Vehicles	3650	1	0	0
	366	Manufacturing of Automotive Bodies and Trailers	3660	1	0	0
	367	Manufacturing of Automotive Parts and Accessories	3670	1	0	0

The econometric model controls for city-year fixed effects μ_{it} and gb4 industry-year fixed effects τ_{ft} , thereby accounting for all time-varying characteristics at both the city and gb4 industry levels, such as demand fluctuations at the city level and market structure changes at the gb4 industry level. Additionally, the city-gb4 industry fixed effects θ_{if} control for all time-invariant characteristics at the city-gb4 industry level. These extensive fixed effects allow the econometric model to robustly control for many omitted variables. Regarding control variables, it is possible that cities-industries with R&D

center foreign firm entry might also be preferred destinations for other types of R&D centers. Consequently, the observed innovation spillovers from R&D center foreign firms could be due to the attractiveness of the city-industry to R&D centers in general. Similarly, cities-industries with R&D center foreign firm entry might also be preferred destinations for other types of foreign enterprises, and the observed innovation spillovers could be attributed to the city-industry's appeal to foreign enterprises. Therefore, this study controls for the logarithm of the number of existing R&D centers at the city-gb2 industry level *lnrdcnum*, and the logarithm of the number of existing foreign enterprises at the city-gb2 industry level *lnforeignnum*, as calculated from the comprehensive registration data of foreign-invested enterprises.

Appendix Table A2: Descriptive statistics of core variables

Variable	Meaning	Observations	Mean	S.D.	Min	Max
<i>lninnovation</i>	City-industry level innovation index	3,968,244	0.0119	0.0784	0	4.5416
<i>hsr×frdc</i>	HSR opened × R&D center foreign firms entry	3,968,244	0.0245	0.1544	0	1
<i>subway×frdc</i>	Subway opened × R&D center foreign firms entry	3,968,244	0.0139	0.1170	0	1
<i>frdc</i>	A binary variable indicating the presence of R&D center foreign firms	3,968,244	0.0381	0.1915	0	1
<i>hsr</i>	A binary variable indicating whether high-speed rail is operational	3,968,244	0.3495	0.4768	0	1
<i>subway</i>	A binary variable indicating whether the subway is operational	3,968,244	0.0567	0.2312	0	1
<i>lnforeignnum</i>	The logarithm of the number of active foreign enterprises	3,968,244	0.9628	1.2509	0	8.0272
<i>lnrdcnum</i>	The logarithm of the number of active R&D center enterprises	3,968,244	0.2238	0.5386	0	5.0876

Appendix B

We argue that " R&D center foreign firms have weaker market competition effects and stronger technological spillover effects, which helps us identify more pure technological spillover effects." To support this claim, we provide some evidence and discuss the potential drawbacks of this approach (identifying and utilizing R&D center foreign firms) in the main text. The evidence we provide is as follows:

First, to the best of our knowledge, the China Industrial Enterprise Database and the China Taxation Survey Database are two large enterprise databases that provide annual financial information for firms. The China Industrial Enterprise Database covers all state-owned enterprises in industrial sectors, as well as non-state-owned enterprises above a certain size (with the size threshold set at 5 million RMB prior to 2011 and at 20 million RMB thereafter). Unfortunately, the available China Industrial Enterprise Database only covers the period from 1998 to 2013, and the data quality of the database deteriorates after 2007. On the other hand, the China Taxation Survey Database is a survey database that includes all industries, but each year a large number of enterprises enter and exit the survey scope. The currently available China Taxation Survey Database only covers the period from 2007 to 2016.

We matched the R&D center foreign firms with the China Industrial Enterprise Database and the China Taxation Survey Database using enterprise names. In this process, to include as many R&D center foreign firms as possible, we did not impose the restriction that R&D center foreign firms must have non-zero insurance participants, ensuring that the analysis results represent an upper bound. The results show that, from 2004 to 2013, 985 industrial sector R&D center foreign firms were matched with 562 enterprises in the China Industrial Enterprise Database (from a total of 775,955 firms), which matched 2,893 firms from a sample size of 3,635,177 in the database. From 2007 to 2016, 2,196 R&D center foreign firms were successfully matched with 1,123 enterprises in the China Taxation Survey Database (from a total of 2,663,814 firms), which matched 4,527 firms from a sample size of 7,093,828 in the database.

To mitigate the potential errors in strict matching based on enterprise names, we employed fuzzy matching in this study, keeping only the results with the highest matching scores, and requiring a matching score of at least 0.99. The results indicate that, from 2004 to 2013, 985 industrial sector R&D center foreign firms were fuzzy matched with 772 enterprises in the China Industrial Enterprise Database (from a total of 775,955 firms), which matched 5,017 firms from a sample size of 3,635,177 in the database. From 2007 to 2016, 2,196 R&D center foreign firms were fuzzy matched with 1,477 enterprises in the China Taxation Survey Database (from a total of 2,663,814 firms), which matched 5,546 firms from a sample size of 7,093,828 in the database.

Using the annual operational data provided by the China Industrial Enterprise Database and the China Taxation Survey Database, we calculated the proportion of R&D center foreign firms in the total scale of firms in the same city-industry (gb2). Additionally, we reported the proportion of other foreign enterprises in these sectors. Panel A of Appendix B1 provides the results from the China Industrial Enterprise Database. As seen, when measuring firm size by production, sales, assets, and employment, the proportion of R&D center foreign firms in the overall firm scale is quite small, with the average proportion only around 0.1%. In contrast, the proportion of other foreign enterprises in the overall firm scale is much larger, significantly higher than the proportion of R&D center foreign firms, with an average proportion of about 7%. If we use fuzzy matching, the proportion of R&D center foreign firms increases slightly, but the average proportion remains around 0.12%.

Panel B of Appendix B1 provides the results from the China Taxation Survey Database. Similarly, when measuring firm size by production, sales, assets, and employment, the proportion of R&D center foreign firms in the overall firm scale is very small, with the average proportion only around 0.07%. In contrast, the proportion of other foreign enterprises in the overall firm scale is significantly larger, with

an average proportion of around 6%. Even with fuzzy matching, the proportion of R&D center foreign firms increases slightly, but the average proportion still remains below 0.08%.

Second, the R&D center foreign firms identified in this study consist of two types: one where the R&D center operates as an independent enterprise, and another where the R&D center is part of a larger corporate structure. In China, each registered enterprise is assigned a unique identification code (Social Unified Credit Code), ensuring that every enterprise operates as an independent legal entity with its own legal person, financial system, and production-sales system. The R&D center foreign firms identified in this study are all independent entities, each possessing its own Social Unified Credit Code. While some of these R&D centers may be subsidiaries of larger groups, they operate independently in practice. For example, Toyota is a well-known global Japanese automobile company. In 2001, it established its Chinese headquarters, ****Toyota Motor (China) Investment Co., Ltd.**** (Social Unified Credit Code: 91110000710938183F), primarily engaged in vehicle production, investment, imports, and distribution in China. In 2011, Toyota set up a dedicated R&D center, ****Toyota Motor Engineering Research & Development Center (China) Co., Ltd.**** (Social Unified Credit Code: 9132000056294875X8), which focuses on research and development, technology transfer, and the commercialization of technological products. Nestlé, a globally recognized Swiss food and beverage company, founded ****Nestlé (China) Co., Ltd.**** in 1995 (Social Unified Credit Code: 9111000062591083XY), with a business scope including food processing and sales. Additionally, Nestlé established independent companies in several provinces and cities across China, responsible for food production and sales. In 2011, Nestlé set up a dedicated R&D center, ****Nestlé R&D Center Shanghai Co., Ltd.**** (Social Unified Credit Code: 913100006074276411), which focuses on research, development, and experimentation, including basic research and product development. This distinction ensures clarity about the independent nature of these R&D centers, separate from the broader corporate structures of their parent companies.

Third, some of the R&D center foreign firms identified in this study are part of larger enterprises, where the R&D center is included within the company structure. For example, ****Oracle****, a globally recognized American software company, established ****Oracle Information Technology Services (Shenzhen) Co., Ltd.**** in China in 2001 (Social Unified Credit Code: 91440300738825551J), primarily engaged in the research and development, production, transfer, technical services, testing, and consulting of computer software. Similarly, ****Siemens****, a well-known German electronics company, set up ****Siemens Electric Co., Ltd. Suzhou**** in China in 1994 (Social Unified Credit Code: 9132050560819750XQ), focusing on the R&D, production, and sales of power electronic devices, as well as providing technical services and consulting.

Fourth, the R&D center foreign firms identified in this study include both purely R&D-focused enterprises and firms that integrate production, sales, and R&D. The former does not generate potential market competition effects, while the latter may produce both market competition effects and technology spillover effects. However, compared to general foreign firms, their technology spillover effects are stronger due to the presence of internal R&D departments rather than focusing solely on production and sales. The regression results of this study provide suggestive evidence for this distinction (e.g., Columns (2) and (4) in Table 1). Specifically, we find that the regression coefficient of the control variable, the logarithm of the number of incumbent foreign firms at the city-gb2 industry level ($\ln foreignnum$), is not statistically significant, and its absolute magnitude is very small. A possible explanation is that the positive impact of technology spillovers from general foreign investments is offset by the negative impact of market competition effects. In contrast, the regression coefficient of the control variable, the logarithm of the number of incumbent R&D center firms at the city-GB2 industry level ($\ln rdcnum$), is significantly positive. This suggests the effectiveness of our approach to identifying R&D center firms—unlike general firms, R&D center firms significantly enhance innovation levels in the same city-industry and generate notable technology spillover effects.

Appendix Table B1: Proportion of R&D center foreign firms and other foreign firms in the same city-gb2 industry (%)

	Strict matching		Fuzzy matching	
	frdc	other foreign firms	frdc	other foreign firms
Panel A	China Industrial Enterprises Database			
production	0.098	7.293	0.121	7.286
sell	0.100	7.220	0.123	7.213
asset	0.090	7.848	0.117	7.841
employment	0.061	6.485	0.079	6.479
Panel B	China Tax Survey Database			
production	0.074	6.597	0.079	6.596
sell	0.072	6.396	0.082	6.393
asset	0.066	6.110	0.076	6.107
employment	0.059	5.150	0.067	5.147

We provided a simple statistical analysis. To include as many R&D center foreign firms as possible, we relaxed the restriction requiring positive insurance participation for these firms to ensure the results represent an upper bound. The data show that among the 496,446 foreign firms established between 2004 and 2020, R&D center foreign firms accounted for 1,513 firms, representing only 0.3% of the total. While the proportion of R&D center foreign firms is small, there is pronounced heterogeneity in their industry distribution. We have included the industry distribution of R&D center foreign firms, grouped according to the high-tech industry classification provided by China's National Bureau of Statistics. Specifically, we calculated the Revealed Comparative Advantage (RCA) index for high-tech and non-high-tech industries:

$$RCA_i = \frac{frdc_i / frdc}{fnum_i / fnum}$$

Where i represents either the high-tech or non-high-tech industry, $frdc_i$ is the number or registered capital of R&D center foreign firms in industry i , $frdc$ is the total number or registered capital of R&D center foreign firms, $fnum_i$ is the number or registered capital of foreign firms in industry i , $fnum$ is the total number or registered capital of foreign firms.

The results are presented in Appendix B2. We found that, based on either firm numbers or registered capital, the RCA of R&D center foreign firms in high-tech industries is significantly higher than in non-high-tech industries, approximately 2.2 to 2.8 times greater. This indicates that R&D center foreign firms are more concentrated in high-tech industries than general foreign firms. We conclude that, while R&D center foreign firms represent a small but meaningful proportion of foreign firms in China, they are especially concentrated in high-tech industries, where they may exert stronger technological spillover effects.

Appendix Table B2: RCA of R&D center foreign firms in (Non-) high-tech industry

RCA	Number	Registered capita
high-tech industry	1.838	2.229
Non-high-tech industry	0.832	0.789

We discuss the differences between the treatment and control groups. First, we examine the differences in innovation levels between the two groups at the city-gb4 industry level, comparing the mean values of the logarithm of the innovation index. The results are presented in Appendix Table B3. We find that, in unconditional comparisons, the innovation index in the treatment group is significantly higher than in the control group. Next, we conduct a panel regression at the city-gb4 industry level, controlling for all variables and fixed effects specified in Equation (1), and examine the residual differences between the treatment and control groups. After accounting for control variables and fixed effects, the innovation index in the treatment group is either no longer significantly higher than in the control group or shows a difference that is over 10 times smaller than in the unconditional comparison.

We then analyze the differences in city-level market size between the treatment and control groups. Here, the treatment and control groups are defined based on the presence of high-speed rail or metro systems, and market size is measured using the logarithms of city GDP and total population. Unconditional comparisons indicate that the market size of cities in the treatment group is significantly larger than in the control group. Conducting a panel regression at the city level, controlling for city and year fixed effects, and examining residual differences, we find that after accounting for fixed effects, the market size of the treatment group is either no longer significantly larger than the control group or shows a difference over 18 times smaller than in the unconditional comparison.

These results suggest that while the treatment group exhibits significantly higher innovation indices and market sizes compared to the control group, these differences are no longer significant or are greatly diminished after controlling for the variables and fixed effects used in this study. To further examine whether the differences in innovation indices and market sizes between the treatment and control groups interfere with the study's estimates, we provide robust econometric evidence in the robustness checks section, as detailed in Section 4.4.

In the regression specified in Equation (2), we further control for the interaction terms between the logarithm of the initial innovation index at the city-gb4 industry level ($\ln innovation_{2003}$) and year fixed effects, which allows for the differential impact of initial innovation levels at the city-industry level across different years. Additionally, we include interaction terms between the logarithms of initial city GDP ($\ln gdp_{2003}$) and total population ($\ln pop_{2003}$) with the industry-year fixed effects, allowing for the differential impact of initial market size at the city level across different industry-year combinations. The initial year is set as 2003, the year before the study period. Appendix Table B3 demonstrates that under these additional controls, the core estimated effects of this study remain robust. Therefore, we conclude that after accounting for the differences in innovation index and city market size between the treatment and control groups, the core estimated effects of this study are still reliable.

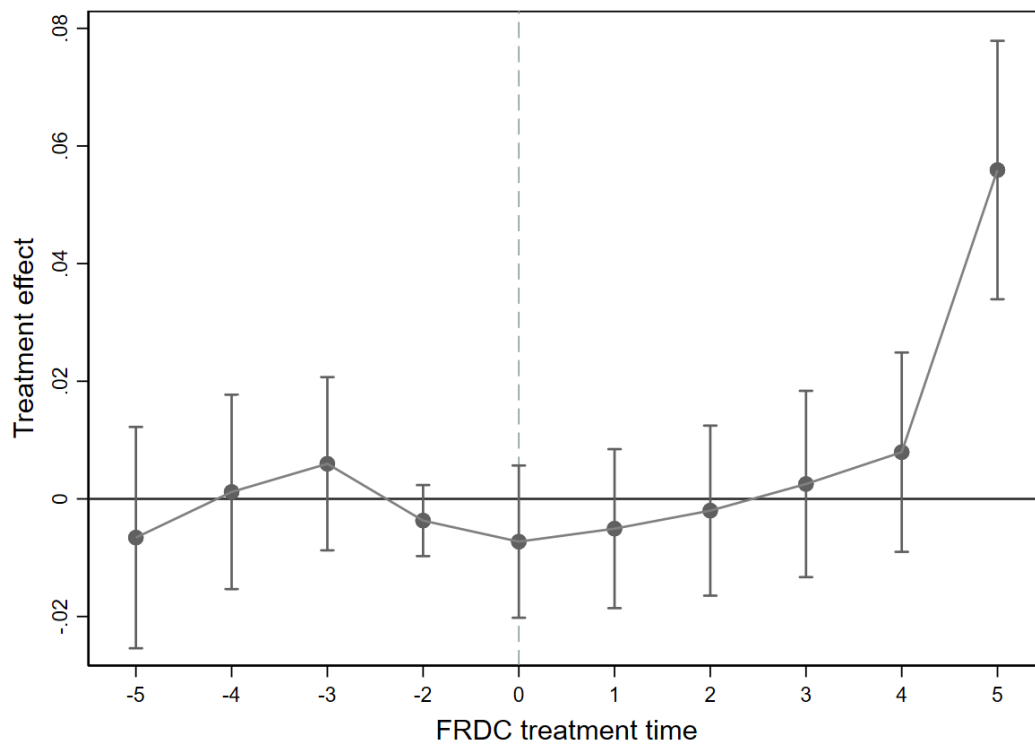
Panel A of Appendix Table B3 provides the results for five different treatment and control group definitions: $hsr \times frdc$ (0 or 1), $subway \times frdc$ (0 or 1), $frdc$ (0 or 1), hsr (0 or 1), $subway$ (0 or 1). Panel B of Appendix Table B3 provides results for two treatment and control group definitions: hsr (0 or 1), $subway$ (0 or 1). In both panels, "0" represents the control group, which indicates no high-speed rail, no metro, or no R&D center foreign firms, while "1" represents the treatment group, indicating the presence of high-speed rail, subway, or R&D center foreign firms.

Appendix Table B3: Difference test between treatment group and control group

panel A		lninnovation			lninnovation residual		
city-gb4 industry level		1	0	diff	1	0	diff
<i>hsr</i> × <i>frdc</i>		0.172	0.008	0.165***	0.015	0.000	0.016***
<i>subway</i> × <i>frdc</i>		0.264	0.008	0.255***	0.017	0.000	0.017***
<i>frdc</i>		0.117	0.008	0.110***	0.001	0.000	0.001***
<i>hsr</i>		0.030	0.002	0.028***	0.000	0.000	0.000
<i>subway</i>		0.123	0.005	0.118***	0.000	0.000	0.000
panel B		lngdp			lngdp residual		
city level		1	0	diff	1	0	diff
<i>hsr</i>		16.369	15.370	0.999***	0.000	0.000	-0.001
<i>subway</i>		17.778	15.596	2.182***	0.022	-0.001	0.023***
panel C		lnpop			lnpop residual		
city level		1	0	diff	1	0	diff
<i>hsr</i>		6.070	5.726	0.344***	0.006	-0.003	0.010***
<i>subway</i>		6.898	5.783	1.115***	0.056	-0.003	0.060***

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix C



Appendix Fig. C1: Results of the event study of R&D center foreign firms entry

Note: Fig. C1 presents the coefficient estimates and corresponding 95% confidence intervals from the event study of R&D center foreign firms entry.

Appendix Table C1: Results of technology spillover effect form R&D center foreign firms in different groups

<i>lninnovation</i>	(1)	(2)	(3)	(4)	(5)
	all	With HSR	Without HSR	With subway	Without subway
<i>frdc</i>	0.0206*** (0.00767)	0.0172* (0.0101)	0.00408 (0.00277)	0.00312 (0.0242)	0.00715** (0.00286)
<i>lnforeignnum</i>	-0.000319 (0.000541)	-0.00455*** (0.00134)	0.000126 (0.000134)	0.00286 (0.00952)	5.01e-05 (0.000197)
<i>lnrdcnum</i>	0.0167*** (0.00219)	0.0110** (0.00532)	0.00203*** (0.000412)	-0.00646 (0.0175)	0.00765*** (0.00101)
City-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes
City-year fixed effects	Yes	Yes	Yes	Yes	Yes
Year-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	3,968,244	1,382,836	2,580,536	221,676	3,741,696
R ²	0.768	0.906	0.770	0.947	0.668

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table C2: Robustness check results (I)

<i>lninnovation</i>	(1)	(2)	(3)	(4)	(5)	(6)
<i>hsr×frdc</i>	0.0124*** (0.000864)	0.0486*** (0.00610)	0.0288*** (0.00587)	0.0458*** (0.00544)	0.0318*** (0.00546)	0.0258*** (0.00511)
<i>subway×frdc</i>	0.0134*** (0.00154)	0.0593*** (0.00781)	0.0643*** (0.00715)	0.0720*** (0.00801)	0.0711*** (0.00724)	0.0664*** (0.00649)
<i>frdc</i>	-0.00629*** (0.00131)	-0.0240*** (0.00730)	-0.0244** (0.00948)	-0.0183*** (0.00702)	-0.0121* (0.00629)	-0.00862 (0.00596)
<i>lnforeignnum</i>	0.000196* (0.000118)	0.000211 (0.000471)	-0.00127** (0.000580)	0.000610 (0.000470)	0.000226 (0.000483)	-0.000507 (0.000551)
<i>lnrdcnum</i>	0.00603*** (0.000513)	0.0147*** (0.00199)	0.0159*** (0.00268)	0.0133*** (0.00172)	0.0136*** (0.00190)	0.00947*** (0.00183)
City-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
City-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	×
provincial-gb4 industry-year fixed effects	×	×	×	×	×	Yes
Observations	3,921,785	4,651,948	3,045,000	4,634,084	3,680,796	3,898,412
R ²	0.720	0.768	0.852	0.723	0.736	0.779

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The binary variables for high-speed rail opening *hsr* and subway opening *subway* are city-level time-varying variables, which are absorbed by the city-year fixed effects.

Appendix Table C3: Robustness check results (II)

<i>lninnovation</i>	(1)	(2)	(3)	(4)	(5)	(6)
<i>hsr×frdc</i>	0.0655*** (0.00760)	0.477*** (0.111)	0.0593*** (0.00777)	0.0477*** (0.0127)	0.0477*** (0.00877)	0.0477*** (0.0135)
<i>subway×frdc</i>	0.0626*** (0.00817)	0.284*** (0.0747)	0.0741*** (0.0100)	0.0581*** (0.0124)	0.0581*** (0.00787)	0.0581*** (0.0127)
<i>frdc</i>	-0.0124 (0.0116)	-0.215*** (0.0811)	-0.0295*** (0.00936)	-0.0238** (0.0102)	-0.0238*** (0.00756)	-0.0238** (0.0105)
<i>lnforeignnum</i>	-0.000240 (0.000512)	-0.00591 (0.0112)	-0.000161 (0.000646)	-0.000102 (0.000561)	-0.000102 (0.00111)	-0.000102 (0.000715)
<i>lnrdcnum</i>	0.0105*** (0.00202)	0.0774*** (0.0230)	0.0161*** (0.00254)	0.0137*** (0.00266)	0.0137*** (0.00366)	0.0137*** (0.00305)
City-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
City-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,968,244	3,968,244	3,968,244	3,968,244	3,968,244	3,968,244
R ²	0.776	0.589	0.757	0.772	0.772	0.772

Note: The values in parentheses in column (1) to column (3) represent city-gb2 industry-level clustered robust standard errors. The values in parentheses in column (4) to column (6) represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The binary variables for high-speed rail opening *hsr* and subway opening *subway* are city-level time-varying variables, which are absorbed by the city-year fixed effects.

Appendix Table C4: Robustness check results (III)

<i>lninnovation</i>	(1)	(2)	(3)
<i>hsr</i> × <i>frdc</i>	0.0382*** (0.00513)	0.0298*** (0.00479)	0.0453*** (0.00576)
<i>subway</i> × <i>frdc</i>	0.0569*** (0.00711)	0.0320*** (0.00578)	0.0524*** (0.00719)
<i>frdc</i>	-0.0185*** (0.00654)	-0.0159*** (0.00536)	-0.0225*** (0.00663)
<i>hsr</i>			
<i>subway</i>			
<i>lnforeignnum</i>	-8.77e-05 (0.000487)	-0.000680 (0.000487)	-0.000440 (0.000549)
<i>lnrdcnum</i>	0.0135*** (0.00187)	0.00338** (0.00157)	0.0106*** (0.00189)
Constant	0.00795*** (0.000550)	0.0115*** (0.000622)	0.00918*** (0.000628)
City-gb4 industry fixed effects	Yes	Yes	Yes
City-year fixed effects	Yes	Yes	Yes
Year-gb4 industry fixed effects	Yes	Yes	Yes
<i>lninnovation2003</i> × year fixed effects	Yes		
<i>lngdp2003</i> × Year-gb4 industry fixed effects		Yes	
<i>lnpop2003</i> × Year-gb4 industry fixed effects			Yes
Observations	3,968,244	3,871,616	3,871,616
R ²	0.805	0.846	0.792

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table C5: Goodman-Bacon decomposition results for equation (1)

Decomposition type	Average treatment effect	weight
Early_v_Late	0.010627	0.00178
Late_v_Early	-0.02733	0.00614
Always_v_timing	-0.06739	0.02544
Never_v_timing	0.08054	0.96664

Note: Appendix Table C4 shows Goodman-Bacon decomposition results for variable *frdc* in equation (1).

Appendix Table C6: Goodman-Bacon decomposition results for equation (2)

<i>hsr×frdc</i>			<i>subway×frdc</i>		
Decomposition type	Average treatment effect	weight	Decomposition type	Average treatment effect	weight
Early_v_Late	0.056513	0.00568	Early_v_Late	0.046296	0.00517
Late_v_Early	-0.10087	0.00648	Late_v_Early	-0.088367	0.00435
Always_v_timing			Always_v_timing	-0.25037	0.00440
Never_v_timing	0.12367	0.98785	Never_v_timing	0.17140	0.98608

Note: Appendix Table C5 shows the Goodman-Bacon decomposition results for variable *hsr×frdc* and *subway×frdc* in equation (2).

Appendix Table C7: Heterogeneous treatment effect estimated results

	(1)	(2)	(3)
<i>frdc</i>	0.021* (0.0107)		
<i>hsr×frdc</i>		0.0769*** (0.0067)	
<i>subway×frdc</i>			0.0757*** (0.0189)

Note: Appendix Table C6 shows the heterogeneous treatment effect estimated results using Callaway & Sant'Anna (2021)'s method. The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix D

Appendix Table D1: Industry heterogeneity results

<i>lninnovation</i>	(1)	(2)	(3)	(4)	(5)	(6)
	High-tech industries			Strategic emerging industries		
Interaction variable Δ	gb4 definition	gb3 definition	gb2 definition	gb4 definition	gb3 definition	gb2 definition
$\Delta \times hsr \times frdc$	0.114*** (0.0239)	0.0949*** (0.0185)	0.0742*** (0.0100)	0.0876*** (0.00962)	0.0638*** (0.00841)	0.0531*** (0.00724)
$\Delta \times subway \times frdc$	-0.0194 (0.0207)	-0.0143 (0.0179)	0.0203 (0.0132)	0.0354*** (0.0105)	0.0523*** (0.0117)	0.0705*** (0.0162)
$hsr \times frdc$	0.0263*** (0.00422)	0.0239*** (0.00404)	-0.00426 (0.00555)	-0.00462 (0.00424)	-0.00559 (0.00620)	-0.00213 (0.00496)
$subway \times frdc$	0.0486*** (0.00773)	0.0472*** (0.00801)	0.0110 (0.00910)	0.0155** (0.00639)	-0.00672 (0.00973)	-0.0262* (0.0146)
$\Delta \times hsr$	-0.00511*** (0.00157)	-0.00433*** (0.00131)	-0.00221*** (0.000764)	-0.00207*** (0.000679)	-0.00113** (0.000526)	-0.000364 (0.000515)
$\Delta \times subway$	0.0701*** (0.0102)	0.0679*** (0.00877)	0.0709*** (0.00504)	0.0653*** (0.00395)	0.0551*** (0.00338)	0.0542*** (0.00336)
$\Delta \times frdc$	0.00273 (0.0210)	0.00555 (0.0169)	-0.00541 (0.0114)	-0.0178* (0.00976)	-0.00642 (0.0115)	0.0219 (0.0176)
$frdc$	-0.0288*** (0.00602)	-0.0286*** (0.00614)	-0.0171** (0.00843)	-0.0111* (0.00614)	-0.0141 (0.0104)	-0.0402** (0.0165)
$lnforeignnum$	0.000115 (0.000490)	7.73e-05 (0.000490)	-6.16e-05 (0.000498)	-0.000175 (0.000501)	-0.000229 (0.000505)	-0.000317 (0.000508)
$lnrdcnum$	0.0115*** (0.00194)	0.0118*** (0.00192)	0.0112*** (0.00186)	0.0115*** (0.00188)	0.0118*** (0.00190)	0.0123*** (0.00192)
City-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
City-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,968,244	3,968,244	3,968,244	3,968,244	3,968,244	3,968,244
R ²	0.778	0.777	0.779	0.780	0.777	0.775

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The binary variables for high-speed rail opening *hsr*, subway opening *subway*, interaction variable Δ and interaction term $hsr \times subway$, $\Delta \times hsr$ and $\Delta \times subway$ are city (industry)-level time-varying variables, which are absorbed by the city-year fixed effects or year-gb4 fixed effects.

Appendix Table D2: Other industry levels analysis results

<i>lninnovation</i>	(1)	(2)	(3)	(4)	(5)	(6)
	City-gb4 industry			City-gb3 industry		City-gb2 industry
<i>frdc_gb2×hsr</i>	0.0477*** (0.00608)			0.131*** (0.0117)		0.376*** (0.0241)
<i>frdc_gb2×subway</i>	0.0581*** (0.00771)			0.142*** (0.0183)		0.347*** (0.0344)
<i>frdc_gb3×hsr</i>		0.0802*** (0.0111)			0.228*** (0.0210)	
<i>frdc_gb3×subway</i>		0.0826*** (0.0120)			0.227*** (0.0256)	
<i>frdc_gb4×hsr</i>			0.104*** (0.0129)			
<i>frdc_gb4×subway</i>			0.117*** (0.0190)			
<i>frdc_gb2</i>	-0.0238*** (0.00723)			-0.0839*** (0.0178)		-0.208*** (0.0396)
<i>frdc_gb3</i>		-0.0325*** (0.0140)			-0.128*** (0.0286)	
<i>frdc_gb4</i>			-0.0330* (0.0196)			
<i>lnforeignnum</i>		-0.000175 (0.000520)	-0.000193 (0.000527)	-0.000980 (0.000953)	-0.00150 (0.000963)	0.000857 (0.00239)
<i>lnrdcnum</i>		0.0177*** (0.00216)	0.0187*** (0.00222)	0.0343*** (0.00474)	0.0453*** (0.00488)	0.162*** (0.0124)
City-gb4 industry fixed effect	Yes	Yes	Yes			
City-year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Year-gb4 industry fixed effect	Yes	Yes	Yes			
City-gb3 industry fixed effect				Yes	Yes	
Year-gb3 industry fixed effect				Yes	Yes	
City-gb2 industry fixed effect						Yes
Year-gb2 industry fixed effect						Yes
Observations	3,968,244	3,968,244	3,968,244	1,299,942	1,299,942	263,898
R ²	0.772	0.771	0.769	0.826	0.823	0.898

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The binary variables for high-speed rail opening *hsr* and subway opening *subway* are city-level time-varying variables, which are absorbed by the city-year fixed effects.